



Planar Kinematics of a Rigid Body

Sliding Rod Example

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MECH 310 - Dynamics May 12, 2026

Review of Previous Lectures

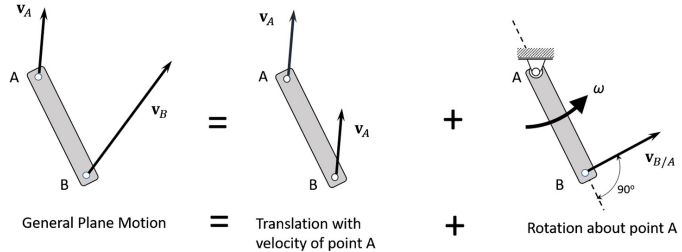
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- geometric constraints,

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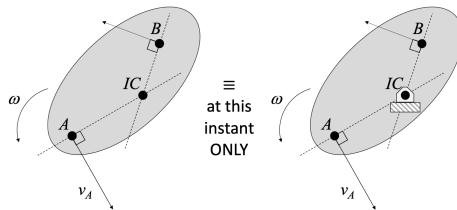
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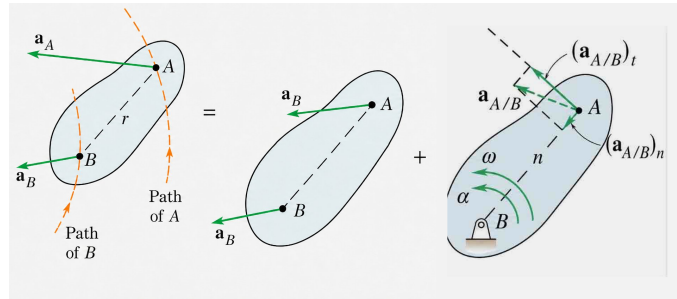
- geometric constraints,
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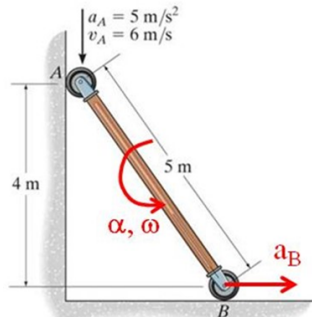


Review of Previous Lectures

In the previous classes, we introduced the main tools for planar rigid-body kinematics:

- geometric constraints,
- relative velocity,
- instantaneous centers of zero velocity,
- relative acceleration.

Known: $\vec{V}_A = -6 \hat{j} \text{ m/s}$, $\vec{a}_A = -5 \hat{j} \text{ m/s}^2$
 $x_B = 3 \text{ m}$, $y_A = 4 \text{ m}$



Today we use one sliding-rod problem to review all four ideas.

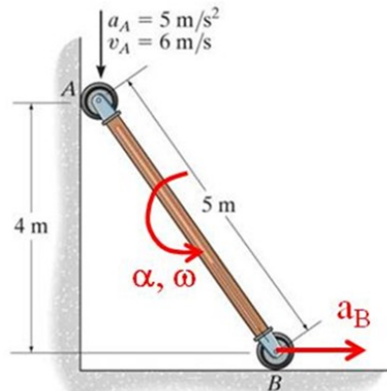
Problem Statement

A 5 m rod has its ends constrained to slide along a vertical wall and a horizontal floor.

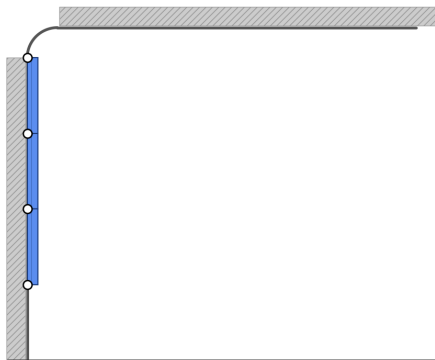
At the instant shown:

- point A moves downward with speed 6 m/s,
- point A has acceleration 5 m/s² downward,
- point B moves horizontally,
- the geometry is a 3-4-5 triangle.

Known: $\vec{V}_A = -6\hat{j}$ m/s, $\vec{a}_A = -5\hat{j}$ m/s²
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Real-World Motivation: Garage Door Motion



Each panel is a rigid bar — both ends constrained to the track



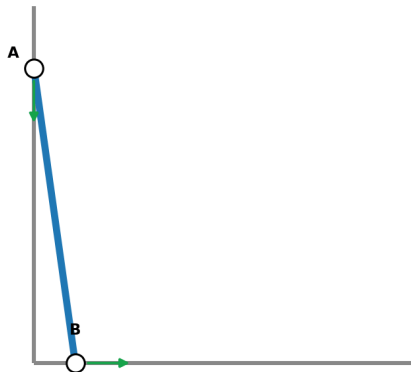
Animation plays in Adobe Acrobat.

A garage door is a familiar example of constrained planar motion.

- Rollers follow the track.
- Each section behaves approximately as a rigid body.
- The motion combines translation and rotation.

This is the same kind of thinking we use for the sliding rod.

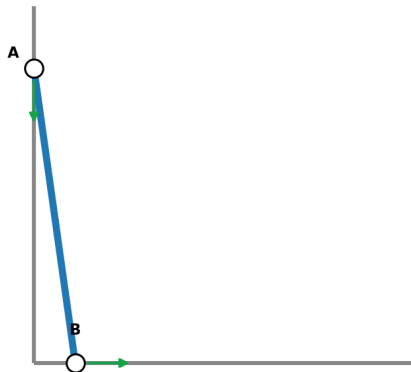
How the Rod Moves



Animation plays in Adobe Acrobat.

- Point A is constrained to move only along the vertical wall.
- Point B is constrained to move only along the horizontal floor.
- Since AB is fixed, the rod translates and rotates at the same time.

How the Rod Moves



Animation plays in Adobe Acrobat.

- Point A is constrained to move only along the vertical wall.
- Point B is constrained to move only along the horizontal floor.
- Since AB is fixed, the rod translates and rotates at the same time.

At the shown instant, the rod forms a 3-4-5 triangle:

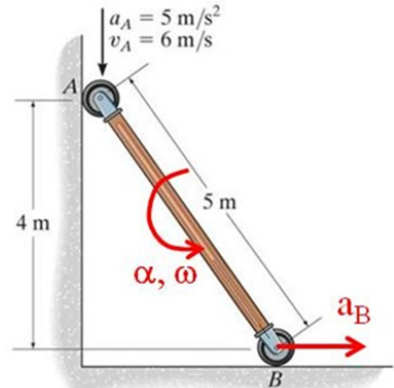
$$x_B = 3 \text{ m}, \quad y_A = 4 \text{ m}, \quad AB = 5 \text{ m}.$$

What We Want to Find

Unknowns

$$\omega, \quad \vec{V}_B, \quad \alpha, \quad \vec{a}_B$$

Known: $\vec{V}_A = -6\hat{j} \text{ m/s}$, $\vec{a}_A = -5\hat{j} \text{ m/s}^2$
 $x_B = 3 \text{ m}$, $y_A = 4 \text{ m}$



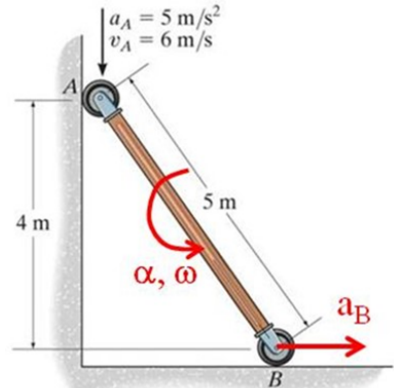
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- **Method 1.** Geometric constraint $\rightarrow \omega, V_B, \alpha, a_B$.

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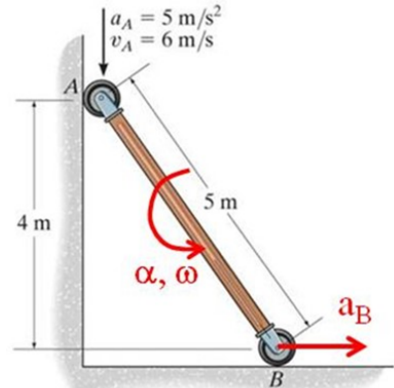
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- **Method 1.** Geometric constraint $\rightarrow \omega, V_B, \alpha, a_B$.
- **Method 2.** Relative velocity $\rightarrow \omega, V_B$.

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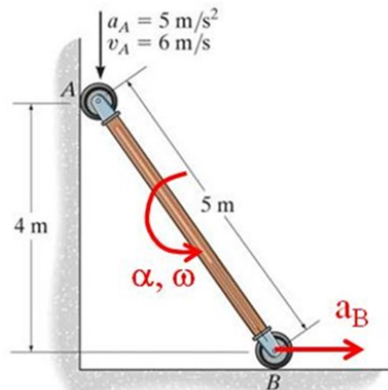
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$$\omega, \quad \vec{V}_B, \quad \alpha, \quad \vec{a}_B$$

- **Method 1.** Geometric constraint $\rightarrow \omega, V_B, \alpha, a_B$.
- **Method 2.** Relative velocity $\rightarrow \omega, V_B$.
- **Method 3.** Instantaneous center $\rightarrow \omega, V_B$.

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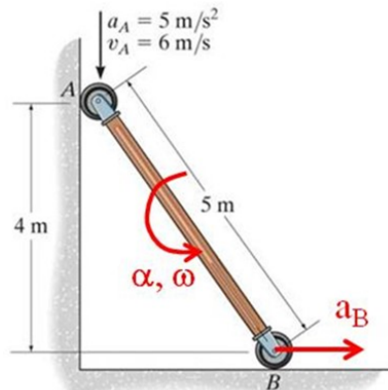
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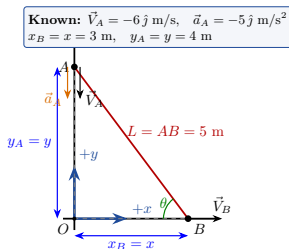
$$\omega, \quad \vec{V}_B, \quad \alpha, \quad \vec{a}_B$$

- **Method 1.** Geometric constraint $\rightarrow \omega, V_B, \alpha, a_B$.
- **Method 2.** Relative velocity $\rightarrow \omega, V_B$.
- **Method 3.** Instantaneous center $\rightarrow \omega, V_B$.
- **Method 4.** Relative acceleration $\rightarrow \alpha, a_B$.

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Method 1: Geometric Constraint



Define the variables from the triangle:

$x = x_B$ = horizontal position of B,

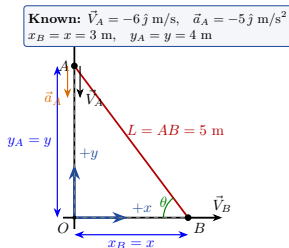
$y = y_A$ = vertical position of A,

$L = AB = 5 \text{ m}$.

Coordinate convention



Method 1: Geometric Constraint



Coordinate convention



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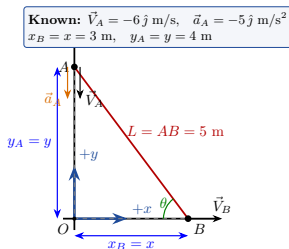
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Since the rod length is fixed,

$$x^2 + y^2 = L^2$$

Method 1: Geometric Constraint



Coordinate convention



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$x = x_B$ = horizontal position of B ,

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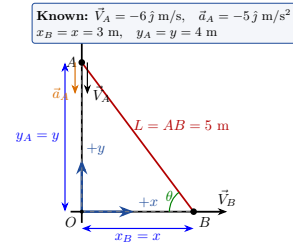
At this instant:

$$x = 3 \text{ m}, \quad y = 4 \text{ m}$$

Velocity From the Constraint

Start with:

$$x^2 + y^2 = L^2$$



Coordinate convention



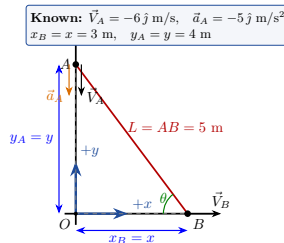
Velocity From the Constraint

Start with:

$$x^2 + y^2 = L^2$$

Differentiate once:

$$2x\dot{x} + 2y\dot{y} = 0 \quad \Rightarrow \quad x\dot{x} + y\dot{y} = 0$$



Coordinate convention



Velocity From the Constraint

Start with:

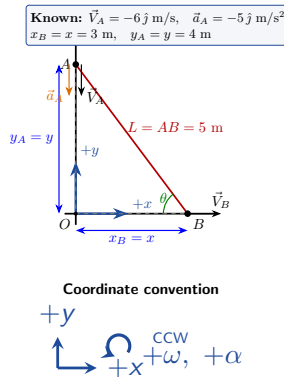
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Since $\dot{y} = V_A = -6 \text{ m/s}$,

$$3\dot{x} + 4(-6) = 0$$



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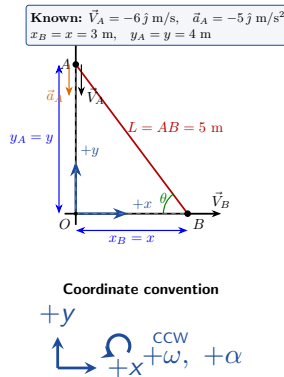
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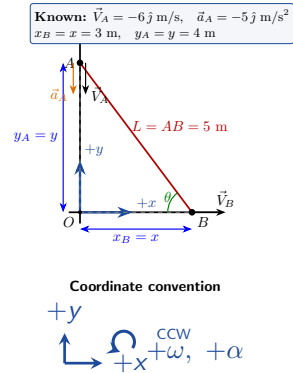
$$\vec{V}_B = \dot{x} \hat{i} = 8 \hat{i} \text{ m/s}$$



Angular Velocity From the Geometry

Use

$$x = L \cos \theta, \quad y = L \sin \theta$$



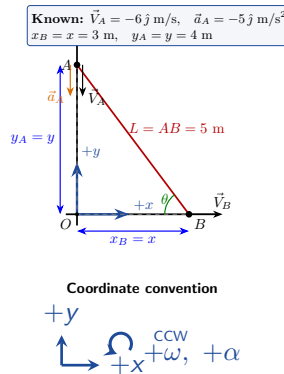
Angular Velocity From the Geometry

Use

$$x = L \cos \theta, \quad y = L \sin \theta$$

Differentiate $y = L \sin \theta$:

$$\dot{y} = L \cos \theta \dot{\theta} = 3\dot{\theta}$$



Angular Velocity From the Geometry

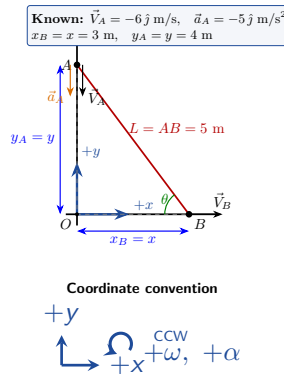
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Differentiate $y = L \sin \theta$:

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$$-6 = 3\dot{\theta} \quad \Rightarrow \quad \dot{\theta} = -2 \text{ rad/s}$$



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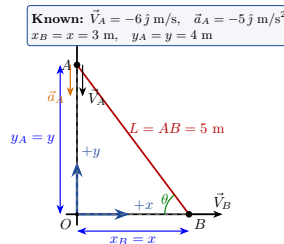
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Note: θ is the rod-floor angle (a parameter), not the rigid-body orientation. The orientation angle of $\vec{r}_{B/A}$ is $-\theta$, so the rigid-body angular velocity is $\omega = -\dot{\theta}$.

$$\omega = -\dot{\theta} = 2 \text{ rad/s CCW}$$



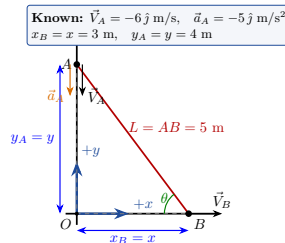
Coordinate convention



Acceleration From the Constraint

Differentiate the constraint a second time:

$$x\ddot{x} + y\ddot{y} + \dot{x}^2 + \dot{y}^2 = 0$$



Coordinate convention



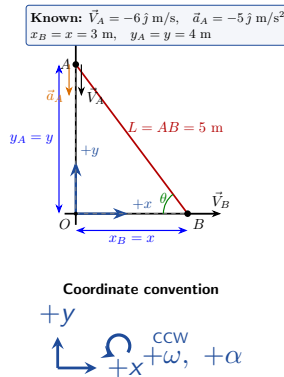
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Substitute the known values:

$$3a_B + 4(-5) + (8)^2 + (-6)^2 = 0$$



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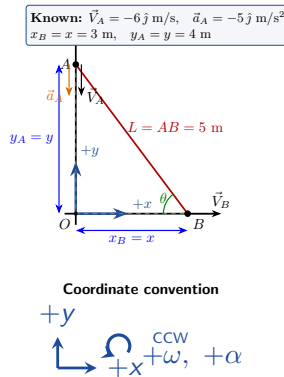
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$$3a_B + 80 = 0$$



Acceleration From the Constraint

Differentiate the constraint a second time:

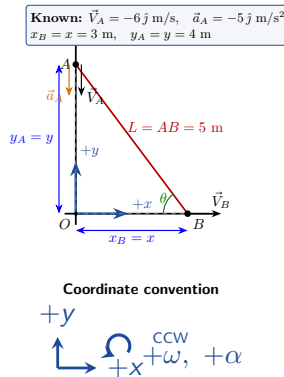
$$x\ddot{x} + y\ddot{y} + \dot{x}^2 + \dot{y}^2 = 0$$

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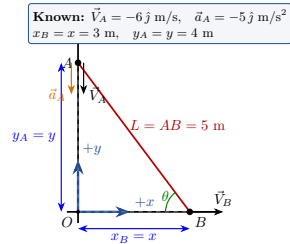
$$\vec{a}_B = -\frac{80}{3} \hat{i} \text{ m/s}^2$$



Angular Acceleration From the Geometry

Differentiate

$$\dot{y} = L \cos \theta \dot{\theta}$$



Coordinate convention

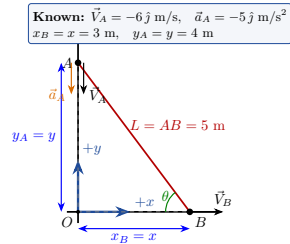


Angular Acceleration From the Geometry

Differentiate

$$\dot{y} = L \cos \theta \dot{\theta}$$

$$\ddot{y} = -L \sin \theta \dot{\theta}^2 + L \cos \theta \ddot{\theta}$$



Coordinate convention



Angular Acceleration From the Geometry

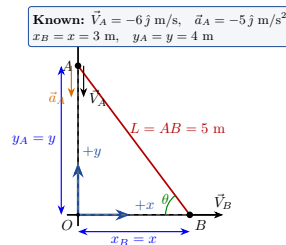
Differentiate

$$\dot{y} = L \cos \theta \dot{\theta}$$

$$\ddot{y} = -L \sin \theta \dot{\theta}^2 + L \cos \theta \ddot{\theta}$$

Using $L \sin \theta = 4$, $L \cos \theta = 3$, $\dot{\theta} = -2$, and $\ddot{y} = -5$:

$$-5 = -4(2)^2 + 3\ddot{\theta}$$



Coordinate convention



Angular Acceleration From the Geometry

Differentiate

$$\dot{y} = L \cos \theta \dot{\theta}$$

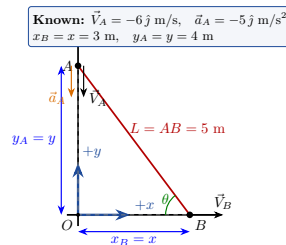
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Using $L \sin \theta = 4$, $L \cos \theta = 3$, $\dot{\theta} = -2$, and $\ddot{y} = -5$:

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$$\ddot{\theta} = \frac{11}{3} \text{ rad/s}^2,$$

$$\alpha = -\frac{11}{3} \text{ rad/s}^2$$



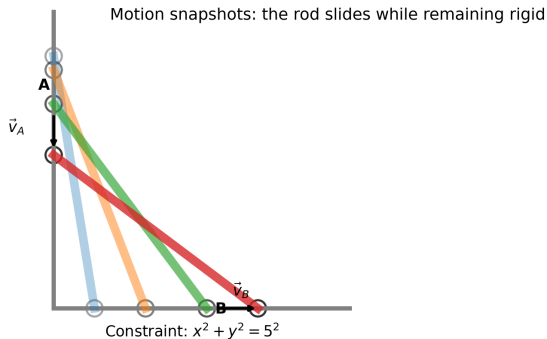
Coordinate convention



Motion Intuition

- The rod translates and rotates at the same time.
- The distance between A and B stays constant.
- The wall and floor impose the motion directions.

$$\vec{r}_{B/A} = 3\hat{i} - 4\hat{j} \text{ m}$$

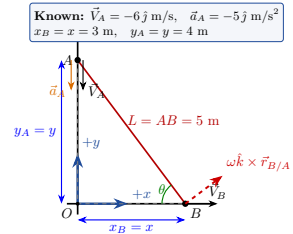


Method 2: Relative Velocity

For two points A and B on the same rigid body:

$$\underset{\substack{D \\ M}}{\vec{V}}_B = \underset{\substack{D \\ M}}{\vec{V}}_A + \underset{\substack{D \\ M}}{\vec{\omega}} \times \underset{\substack{D \\ M}}{\vec{r}}_{B/A}$$

This vector equation gives two scalar component equations, so we can solve for two unknown magnitudes: V_B and ω .



Coordinate convention



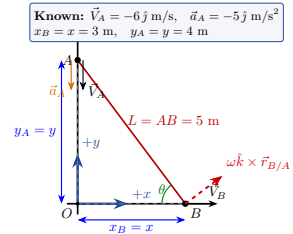
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This vector equation gives two scalar component equations, so we can solve for two unknown magnitudes: V_B and ω .
Substitute the known vectors:

$$V_B \hat{i} = -6\hat{j} + \omega \hat{k} \times (3\hat{i} - 4\hat{j})$$



Coordinate convention



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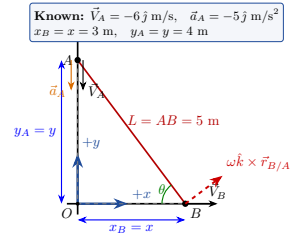
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Since

$$\hat{k} \times (3\hat{i} - 4\hat{j}) = 4\hat{i} + 3\hat{j},$$



Coordinate convention



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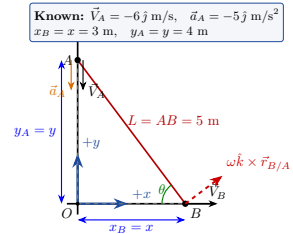
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Substitute the known vectors:

$$V_B \hat{i} = -6\hat{j} + \omega \hat{k} \times (3\hat{i} - 4\hat{j})$$

Since

$$\hat{k} \times (3\hat{i} - 4\hat{j}) = 4\hat{i} + 3\hat{j},$$

$$V_B \hat{i} = -6\hat{j} + \omega(4\hat{i} + 3\hat{j})$$



Coordinate convention

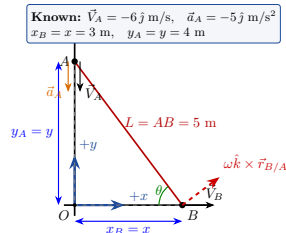


Relative Velocity Result

Equating components in $V_B \hat{i} = -6\hat{j} + \omega(4\hat{i} + 3\hat{j})$:

$$\hat{i}: V_B = 4\omega$$

$$\hat{j}: 0 = -6 + 3\omega$$



Coordinate convention



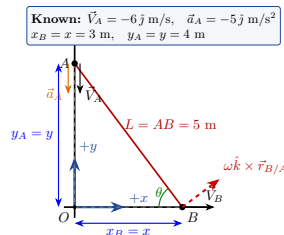
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$$\omega = 2 \text{ rad/s CCW}$$



Coordinate convention



Relative Velocity Result

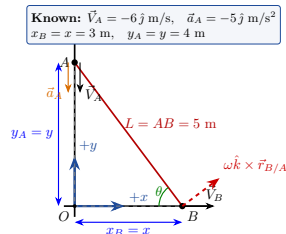
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$$\omega = 2 \text{ rad/s CCW}$$

$$\vec{V}_B = 8\hat{i} \text{ m/s}$$

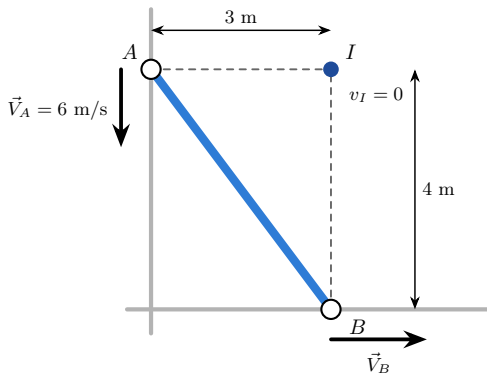


Coordinate convention



Method 3: Instantaneous Center

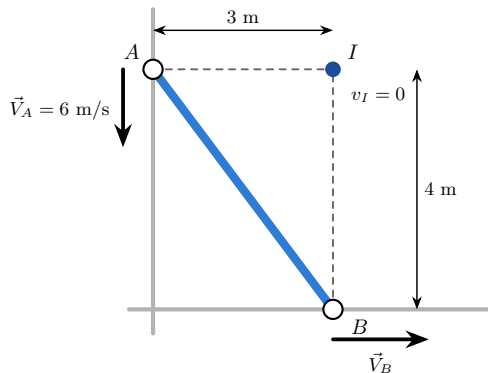
Instantaneous center of zero velocity



- We know the directions of $\vec{V}_A$ and $\vec{V}_B$ from the guides.
- Draw a line through each point perpendicular to its velocity direction.
- At A, $\vec{V}_A$ is vertical, so IA is horizontal.
- At B, $\vec{V}_B$ is horizontal, so IB is vertical.
- The two perpendicular lines intersect at I, the instantaneous center.

Method 3: Instantaneous Center

Instantaneous center of zero velocity



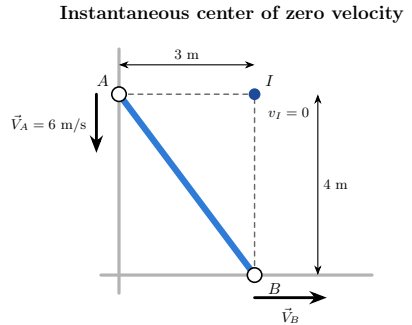
- We know the directions of $\vec{V}_A$ and $\vec{V}_B$ from the guides.
- Draw a line through each point perpendicular to its velocity direction.
- At A, $\vec{V}_A$ is vertical, so IA is horizontal.
- At B, $\vec{V}_B$ is horizontal, so IB is vertical.
- The two perpendicular lines intersect at I, the instantaneous center.

$$IA = 3 \text{ m}, \quad IB = 4 \text{ m}$$

Instantaneous Center Result

The body rotates about I . Since V_A is known, use point A to find ω :

$$V_A = \omega IA$$



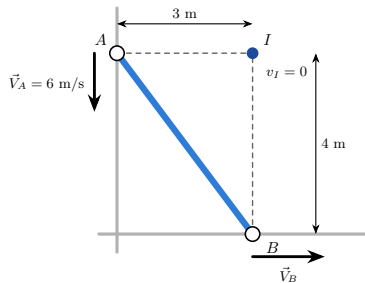
Instantaneous Center Result

The body rotates about I . Since V_A is known, use point A to find ω :

$$V_A = \omega IA$$

$$6 = \omega(3) \quad \Rightarrow \quad \boxed{\omega = 2 \text{ rad/s CCW}}$$

Instantaneous center of zero velocity



Instantaneous Center Result

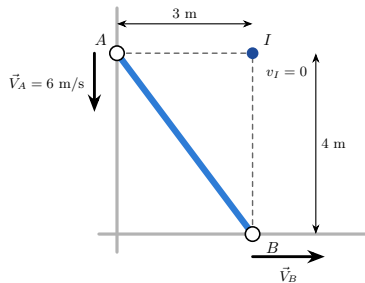
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$$V_B = \omega IB = 2(4) = 8 \text{ m/s}$$

Instantaneous center of zero velocity



Instantaneous Center Result

The body rotates about I . Since V_A is known, use point A to find ω :

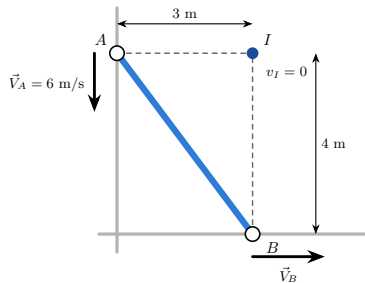
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$$\boxed{\vec{V}_B = 8 \hat{i} \text{ m/s}}$$

Instantaneous center of zero velocity



Instantaneous Center Result

The body rotates about I . Since V_A is known, use point A to find ω :

$$V_A = \omega IA$$

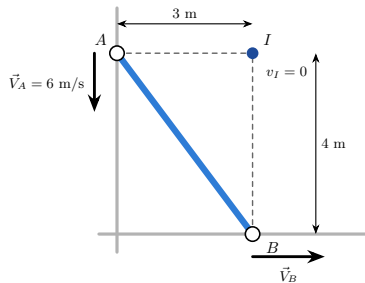
$$6 = \omega(3) \quad \Rightarrow \quad \boxed{\omega = 2 \text{ rad/s CCW}}$$

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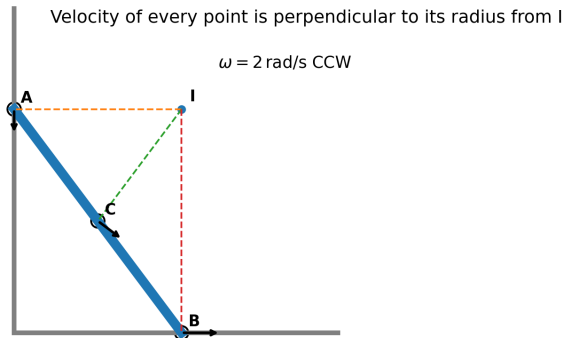
$$\boxed{\vec{V}_B = 8 \hat{i} \text{ m/s}}$$

Cross-check: all three methods give $\omega = 2 \text{ rad/s}$ and $V_B = 8 \text{ m/s}$.

Instantaneous center of zero velocity



Velocity Field Interpretation

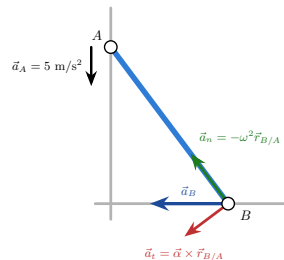


All point velocities are perpendicular to the radius from the instantaneous center.

Method 4: Relative Acceleration

$$\underset{\substack{D \\ M}}{\vec{a}_B} = \underset{\substack{D \\ M}}{\vec{a}_A} + \underset{\substack{D \\ M}}{\vec{\alpha}} \times \underset{\substack{D \\ M}}{\vec{r}_{B/A}} - \omega^2 \underset{\substack{D \\ M}}{\vec{r}_{B/A}}$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action ($\perp \vec{r}_{B/A}$) is known.

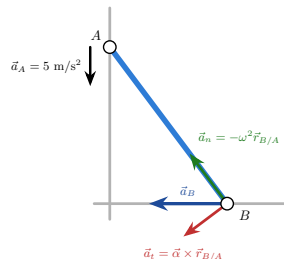
Method 4: Relative Acceleration

$$\underset{\substack{D \\ M}}{\vec{a}}_B = \underset{\substack{D \\ M}}{\vec{a}}_A + \underset{\substack{D \\ M}}{\vec{\alpha}} \times \underset{\substack{D \\ M}}{\vec{r}}_{B/A} - \underset{\substack{D \\ M}}{\omega^2} \underset{\substack{D \\ M}}{\vec{r}}_{B/A}$$

Tangential part:

$$\vec{a}_t = \vec{\alpha} \times \vec{r}_{B/A}$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action ($\perp \vec{r}_{B/A}$) is known.

Method 4: Relative Acceleration

$$\underset{\substack{D \\ M}}{\vec{a}_B} = \underset{\substack{D \\ M}}{\vec{a}_A} + \underset{\substack{D \\ M}}{\vec{\alpha}} \times \underset{\substack{D \\ M}}{\vec{r}_{B/A}} - \underset{\substack{D \\ M}}{\omega^2 \vec{r}_{B/A}}$$

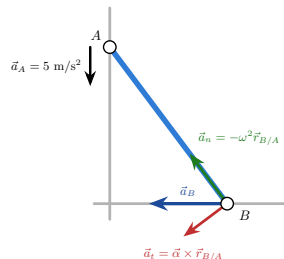
Tangential part:

$$\vec{a}_t = \vec{\alpha} \times \vec{r}_{B/A}$$

Normal part:

$$\vec{a}_n = -\omega^2 \vec{r}_{B/A}$$

Relative acceleration: tangential + normal



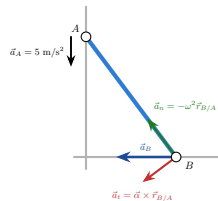
$\vec{a}_t$ shown for the solved α ; a priori only its line of action ($\perp \vec{r}_{B/A}$) is known.

Acceleration Equation

Substitute known values:

$$a_B \hat{i} = -5 \hat{j} + \alpha \hat{k} \times (3 \hat{i} - 4 \hat{j}) - (2)^2 (3 \hat{i} - 4 \hat{j})$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action

($\perp \vec{r}_{B/A}$) is known.

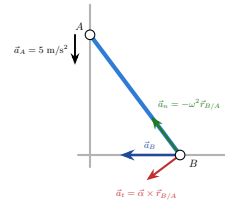
Acceleration Equation

Substitute known values:

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$$a_B \hat{i} = -5\hat{j} + \alpha(4\hat{i} + 3\hat{j}) - 4(3\hat{i} - 4\hat{j})$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action

($\perp \vec{r}_{B/A}$) is known.

Acceleration Equation

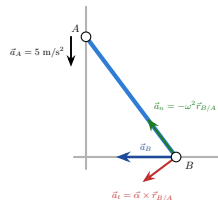
Substitute known values:

$$a_B \hat{i} = -5\hat{j} + \alpha \hat{k} \times (3\hat{i} - 4\hat{j}) - (2)^2(3\hat{i} - 4\hat{j})$$

$$a_B \hat{i} = -5\hat{j} + \alpha(4\hat{i} + 3\hat{j}) - 4(3\hat{i} - 4\hat{j})$$

$$a_B \hat{i} = (4\alpha - 12)\hat{i} + (3\alpha + 11)\hat{j}$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action

($\perp \vec{r}_{B/A}$) is known.

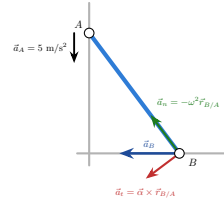
Acceleration Result

Equating components in $a_B \hat{i} = (4\alpha - 12)\hat{i} + (3\alpha + 11)\hat{j}$:

$$\hat{i}: a_B = 4\alpha - 12$$

$$\hat{j}: 0 = 3\alpha + 11 \quad (\text{since } B \text{ stays on the floor})$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action
($\perp \vec{r}_{B/A}$) is known.

Acceleration Result

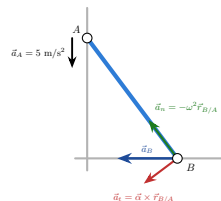
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$$\alpha = -\frac{11}{3} \text{ rad/s}^2$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action
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Acceleration Result

Equating components in $a_B \hat{i} = (4\alpha - 12)\hat{i} + (3\alpha + 11)\hat{j}$:

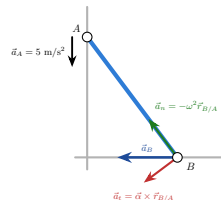
$$\hat{i}: a_B = 4\alpha - 12$$

$$\hat{j}: 0 = 3\alpha + 11 \quad (\text{since } B \text{ stays on the floor})$$

$$\alpha = -\frac{11}{3} \text{ rad/s}^2$$

$$a_B = 4\alpha - 12 = 4\left(-\frac{11}{3}\right) - 12 = -\frac{80}{3} \text{ m/s}^2$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action
($\perp \vec{r}_{B/A}$) is known.

Acceleration Result

Equating components in $a_B \hat{i} = (4\alpha - 12)\hat{i} + (3\alpha + 11)\hat{j}$:

$$\hat{i}: a_B = 4\alpha - 12$$

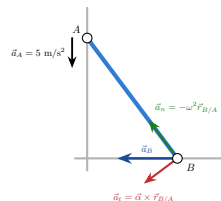
$$\hat{j}: 0 = 3\alpha + 11 \quad (\text{since } B \text{ stays on the floor})$$

$$\alpha = -\frac{11}{3} \text{ rad/s}^2$$

$$a_B = 4\alpha - 12 = 4\left(-\frac{11}{3}\right) - 12 = -\frac{80}{3} \text{ m/s}^2$$

$$\vec{a}_B = -\frac{80}{3} \hat{i} \text{ m/s}^2$$

Relative acceleration: tangential + normal



$\vec{a}_t$ shown for the solved α ; a priori only its line of action
($\perp \vec{r}_{B/A}$) is known.

Summary

$$\omega = 2 \text{ rad/s CCW}$$

$$\vec{V}_B = 8 \hat{i} \text{ m/s}$$

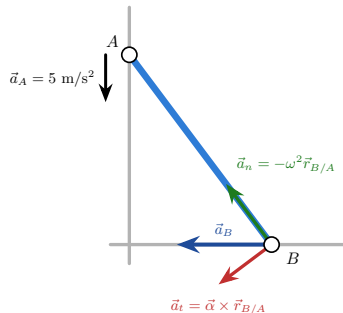
$$\alpha = -\frac{11}{3} \text{ rad/s}^2 = 3.67 \text{ rad/s}^2 \text{ CW}$$

$$\vec{a}_B = -\frac{80}{3} \hat{i} \text{ m/s}^2 = 26.67 \text{ m/s}^2 \text{ left}$$

α is CW (opposite ω) $\Rightarrow$ rod decelerating.

$\vec{a}_B$ points left (opposite $\vec{V}_B$) $\Rightarrow$ B slowing down.

Relative acceleration: tangential + normal



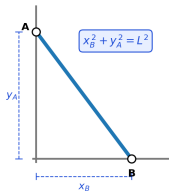
- ω, α : rotation of the rod
- $\vec{V}_B, \vec{a}_B$: motion of B along the floor
- $\vec{a}_n$: centripetal ($B \rightarrow A$)
- $\vec{a}_t$: tangential ($\perp$ rod)

Common Pitfalls

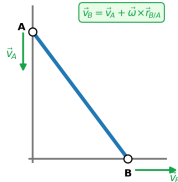
- **Sign of V_A .** “6 m/s downward” means $V_A = -6$ m/s in $\hat{j}$ components, not +6.
- **Velocity vs. acceleration direction.** $\vec{V}$ is the direction of motion; $\vec{a}$ tells you accelerating ($\vec{a}$ along $\vec{V}$) or decelerating ($\vec{a}$ opposite $\vec{V}$). Opposite signs do *not* mean reversal.
- **Parameter θ vs. rigid-body orientation.** If you parametrize with $x = L \cos \theta$, then $\dot{\theta}$ may have the opposite sign to ω . Decide once which θ you mean.
- **Direction of $\vec{a}_t$.** Only its line of action ($\perp \vec{r}_{B/A}$) is known a priori; the sense follows from the sign of α *after* solving.
- **Direction of $\vec{a}_n$.** Always points from B toward A (centripetal).
- **Units check.** $\omega^2 |\vec{r}_{B/A}|$ has units $(\text{rad/s})^2 \cdot \text{m} = \text{m/s}^2$ (radians are dimensionless).

Key Takeaways

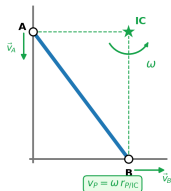
1. Geometric Constraint



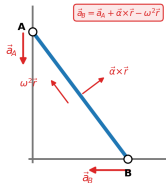
2. Relative Velocity



3. Instantaneous Center of Zero Velocity



4. Relative Acceleration



1 Geometric constraint:

$$x_B^2 + y_A^2 = L^2$$

Direct route to ω , V_B , α , a_B .

2 Relative velocity:

$$\vec{V}_B = \vec{V}_A + \vec{\omega} \times \vec{r}_{B/A}$$

Systematic vector method.

3 Instantaneous center:

$$v_P = \omega r_{P/IC}$$

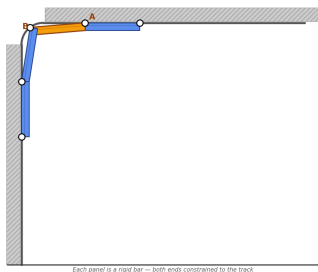
Visual interpretation of velocity.

4 Relative acceleration:

$$\vec{a}_B = \vec{a}_A + \vec{\alpha} \times \vec{r} - \omega^2 \vec{r}$$

Tangential + normal split.

Design Question



Why is the garage-door track curved instead of a sharp corner?

- A sharp corner would change the velocity direction almost instantly.
- That would require very large angular acceleration α .
- Large α means large required torque: $\tau \sim I\alpha$.
- A smooth curve keeps ω , α , roller forces, and motor torque manageable.

Design link: these kinematic calculations help engineers size motors, rollers, and track curvature.

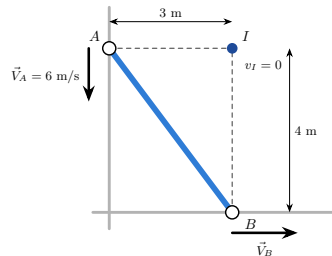
Practice: Theory, Compute and Visualize

As homework, write or complete a Python/MATLAB script that:

- defines the rod length L ,
- varies $y_A(t)$,
- computes

$$x_B(t) = \sqrt{L^2 - y_A^2(t)},$$

- computes $V_B(t)$ and $a_B(t)$,
- plots the rod motion and velocity.



Task

Use the geometric constraint to animate the sliding rod and plot $x_B(t)$, $V_B(t)$, and $a_B(t)$. Then explain where acceleration becomes large and why.

Check the Sign Convention

The first step in any kinematics problem is to define the coordinate system. Signs are not absolute; they come from the positive directions we choose. Do not forget the right-hand rule when defining positive rotation and the $+\hat{k}$ direction.

Student Work

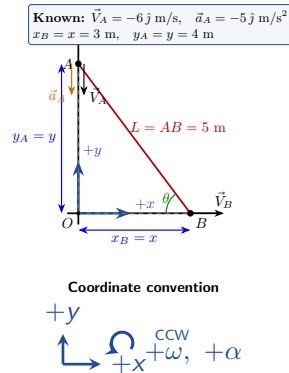
A student used

$$\vec{V}_A = +6\hat{j}$$

and found

$$\vec{V}_B = -8\hat{i}.$$

Why?



Check the Sign Convention

The first step in any kinematics problem is to define the coordinate system. Signs are not absolute; they come from the positive directions we choose. Do not forget the right-hand rule when defining positive rotation and the $+\hat{k}$ direction.

Student Work

A student used

$$\vec{V}_A = +6\hat{j}$$

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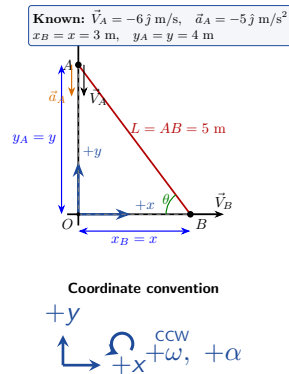
$$\vec{V}_B = -8\hat{i}.$$

Why?

Discussion point: With our coordinate convention, $+\hat{j}$ is upward. Since A is moving downward,

$$\vec{V}_A = -6\hat{j}.$$

The student used the opposite sign, so the computed direction of $\vec{V}_B$ was reversed.



Before You Leave

Answer **one** of these questions in one or two sentences:

- 1 Explain why acceleration of B can point left while velocity of B points right.
- 2 Which method do you trust most for velocity: constraint, relative velocity, or instantaneous center? Why?

Take care, and see you next class.

Next time: rotating coordinate systems and the Coriolis effect.